

WARIT YUVANIYAMA (NOTE)

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Third Year Computer Engineering Student
Sirindhorn International Institute of Technology (SIIT), Thammasat University



SUMMARY

High-achieving Computer Engineering student (3.98 CGPA) and President of SIIT Academy Club (OrcaBOT). Proficient in **embedded systems design** using **C++**, **Python**, and **ROS2**, with peer-reviewed research publications. Proven leader in managing technical procurement (budgeting **100,000+ THB**) and an award-winning communicator with a track record in technical pitching for innovative solutions in Sustainability and Smart Cities.

EDUCATION

Bachelor of Computer Engineering, SIIT, Thammasat University Expected 2027
CGPA: 3.98 (Expected First-Class Honors); Recipient of Academic Outstanding Student Scholarship.

- *Relevant Coursework*: Microcontrollers & Applications, Circuit Analysis, Database Systems, Operating Systems, Computer Architecture, Artificial Intelligence & Applications, Object-Oriented Programming, Algorithm Design.

High School (DPSTE Program), Samsenwittayalai School Graduated Feb 2023
CGPA: 3.96

- *Relevant Coursework*: Computer and Algorithm, Programming, Introduction to Research Methodology, Design and Technology, Science Project, English for Academic Purpose.

SKILLS

Programming	Python, C++, Java, ROS2, micro-ROS, Basic Assembly, LaTeX
Systems & DB	Relational Database Design, SQL (MySQL), Ubuntu Linux
Hardware	ESP32, Raspberry Pi, Arduino, KiCad, Sensors/Actuators, MATLAB
Technical	Project Management, Research & Technical Writing, Technical Pitching
Soft Skills	System Thinking, Creative Problem Solving, Leadership, Cross-cultural Teamwork
Languages	Thai (Native), English (Fluent)
English Scores	TU-GET: 850 — SAT: 1420 (M:790, EBRW:630) — TGAT1: 96.7 — TOEFL (ITP): 553 (B2)

EXPERIENCE

President & Embedded Systems Lead, SIIT Academy Club (OrcaBOT Robotics) Oct 2023 – Present

- **Leadership**: Leading a 30-member club and managing a procurement budget of **100,000+ THB** for motors, sensors, and custom PCB fabrication.
- **Firmware Architecture**: Developed custom C++ libraries for motor control and implemented holonomic drive algorithms; redesigned serial and wireless nRF radio communication systems to significantly increase responsiveness and stability.
- **Systems Integration**: Orchestrating technical collaboration across the High-level, Embedded, Electrical, and Mechanical departments to ensure seamless hardware-software integration for soccer robots.
- **Technical Communication**: Served as the primary author for the Software Architecture section (System Overview, Low-Level, and High-Level Design) of the RoboCup 2024 Team Description Paper, translating complex cross-departmental logic into a peer-reviewed format. [[Link: Team Description Paper \(RoboCup 2024\)](#)]

Intern, NECTEC, NSTDA

May 2024 – Dec 2025

- Implemented Fast Segment Anything Model (FastSAM) for water surface level monitoring system.
- Developed a scaled wave-tank coastal erosion model to test the Water Surface Level Estimation model.
- Presented research findings at international venues, including IUKM 2025 (Vietnam).

Intern, Chemistry Department, Faculty of Science, Mahidol University

Mar 2022 (10 days)

- Assisted in organic chemistry synthesis; achieved a 33.8% yield (2.35g) for naphtho[2,1-b]furan-1(2H)-one.
- Utilized ChemDraw software and performed laboratory troubleshooting.

RESEARCH PUBLICATIONS

- Chetsawang, J., Sricharoenchit, K., Jairak, K., **Yuvaniyama, W.** (*Presenter*), *et al.* (2025). Water Surface Level Estimation Based on the Fast Segment Anything Model for Coastal Monitoring Systems. In: Huynh, V.N., *et al.* (eds) *Integrated Uncertainty in Knowledge Modelling and Decision Making. IUKM 2025*. Lecture Notes in Computer Science, vol 15585. Springer, Singapore. DOI: [10.1007/978-981-96-4606-7_28](https://doi.org/10.1007/978-981-96-4606-7_28)
- Thongsupan, T., Sureechainirun, N., **Yuvaniyama, W.**, *et al.* (2024). OrcaBOT Team Description. *RoboCup 2024 Symposium and World Championship: Team Description Papers - Small Size League*. [\[Semantic Scholar\]](#)

PROJECTS

- **CareAir: Air Quality Monitoring & Carbon Capture System (Hackathon Winner)**
Developed “CareAir”, a proof-of-concept carbon capture machine designed to improve air quality in classrooms/workplaces. Awarded **1st Place** (Physical Wellness Track) at Thammasat Hackathon: Future Wellness 2024.
- **CrystalEyes: Automated Crystal Imaging System**
Developed a low-cost (10k-15k THB) CNC-based digital microscope system to automate crystallization screening, replicating core functionality of commercial systems costing 1M+ THB. Iteratively designed and built from high school through university. Demonstrated with lysozyme growth and MOF studies.
Links: [\(Video\)](#) — [\(Report/Poster\)](#)
- **Autolocate: Condo Parking Management System**
Designed a relational database backend using **SQL and Express.js** to manage high-traffic condo parking operations, implementing a three-tier connection pool system to enforce role-based access control (RBAC) and ensuring secure authentication via JWT and HttpOnly cookies.
Link: [\(GitHub Repository\)](#)
- **Bedroom Tree: Enzymatic Indoor Carbon Capture Concept**
Co-developed concept for an indoor CO₂ capture device using carbonic anhydrase enzyme. Reached semi-final round, Thailand Innovation Award 2023.
Link: [\(Report/Poster\)](#)
- **Thailand CANSAT - Rocket Competition 2022**
Designed mission parameters and internal satellite subsystems for the team’s CANSAT submission.

WORKSHOPS & INTERNATIONAL PROGRAMS

- **Staff, International Project-Based Learning (iPBL) Workshop: Rescue Robot** August 2025
Managed technical logistics and material procurement for international teams from SIIT and OIT Japan. Provided hands-on mentorship in robot design and system debugging, specializing in ROS2, micro-ROS, and Linux integration on Raspberry Pi and ESP32 platforms, while facilitating cross-cultural collaboration between Thai and Japanese students.
- **Participant, 11th Problem-Based Learning (PBL) Workshop, Taipei Tech, Taiwan** August 2024
Collaborated within an international team (12 universities represented) to design and build a smart autonomous vehicle using Arduino, sensors/actuators, and 3D modeling for tasks including color recognition, positioning, and object grasping over an 8-day period.
- **Participant, OIT-SIIT International Project-Based Learning (iPBL), Osaka, Japan** July 2024
Developed a functional mini rescue robot prototype from scratch in 1 week using ROS (on Raspberry Pi) for communication/video streaming and Arduino for motor/arm control; successfully completed simulated rescue mission objectives.

AWARDS AND RECOGNITION

- **1st Place Winner**, Thammasat Hackathon: Future Wellness 2024 (CareAir Team)
- **Finalist**, ASEAN-China-India Youth Leadership Summit 2024 Sustainability Startathon (Singapore)
- **3rd Place (of 200 teams)**, 5th Kibo Robot Programming Challenge Thailand (Plumber Pups Team)
Developed autonomous navigation and **ArUco marker detection** scripts in **Java** for ISS Astrobee robots.
- **Semi-finalist**, Thailand Innovation Award 2023 (Bedroom Tree Project)
- **1st Place (Gold Medal)**, Science Project & Invention, SKIT National Competition 2023
- **Research & Science Project Awards**, 2022 DPSTE Conference (1st Place Poster, 2nd Place Oral)